

PROGRAMMING METHODS - LABORATORY 7

Program 01

Write a program that generates random numbers using the methods listed below. The program should allow:

- selection of one of the methods: Linear Congruential Generator or Additive Congruential Method;
- generation of a sequence of n pseudorandom numbers;
- selection of an integer seed for each method;
- saving the generated sequence of numbers to a text file in column form.

Linear congruential method

$$X_{n+1} = (a * X_n + c) \bmod m$$

- X_{n+1} - the n -th pseudorandom number;
- X_n - the previous pseudorandom number, i.e. the previous result and the internal state of the generator;
- a - multiplier, selected appropriately;
- c - increment, selected appropriately. If $c \neq 0$, it is an additive LCG; if $c = 0$, it is a multiplicative LCG;
- m - modulus, i.e. the range of generated numbers.

The initial value x_0 (seed) is any integer from the range covered by the generator period.

Important note

For the LCG generator, write a program that calculates the coefficients m , a , and c on the basis of the maximum pseudorandom number $X_{\max}$ provided as input.

Additive congruential method

Use recommended pairs of subscript values (l, k) that produce long periods modulo 2. The source sheet includes a table of recommended pairs.

Laboratory example

- Design a generator based on the linear congruential method (LCG) that generates pseudorandom numbers in the interval $<1, 12>$.
- Design a generator based on the additive congruential method that generates pseudorandom numbers for $n = 6$ and $m = 8$.

Additional example - pseudorandom number generation in C++

```
#include <iostream>
#include <random>
using namespace std;

int main() {
    random_device rd;          // starts the pseudorandom number generator
    mt19937 mt(rd());          // Mersenne Twister generator

    // integer number generation
    uniform_int_distribution<int> dist(1, 10); // restricts generation to (1,10)

    // real number generation:
    // uniform_real_distribution<double> dist(1, 10);

    for (int i = 0; i < 16; ++i)
        cout << dist(rd) << "\t" << dist(mt) << "\n";

    cin.ignore();
    getchar();
    return 0;
}
```

The above code starts the pseudorandom number generator called Mersenne Twister (MT). MT is a large shift register with feedback causing rotation and bit mixing. The register length is 19937 bits (a Mersenne

number), which in memory occupies 624 32-bit words.

Literature

Wieczorkowski R., Zieliński R.: Computer Random Number Generators, WN-T, Warsaw 1997. Knuth Donald E.: The Art of Computer Programming, Vol. 2, Seminumerical Algorithms, WN-T, Warsaw 2002.

Submission rules

The .cpp file(s) should be named:

Surname_Name_Program_01.cpp

The source files, all other created files, and test text files should be placed in the folder Surname_Name_Laboratory_7.

The folder should be packed as a .rar or .zip archive and submitted to the course folder for Programming Methods - laboratory 7 on delta.pk.edu.pl.